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The DeGerlia Time Dilation Framework

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ABSTRACT: General Relativity reveals the relationship between space and time that every system in the universe must observe. But above all, it speaks to how systems relate. And describes how systems observed from different frames of reference can appear differently. The relative change in one's frame of reference, or of the observed, will be perceived very differently from the observer than from the observed system itself. In this paper, we discuss the factors at play with relativistic time dilation in a simple, versatile, and predictive manner. We demonstrate that time dilation between two systems and their respective spatial mass distribution by calculating and verifying the relative pace of time between uniform spherical systems from comparative geometry alone. Using the compactness relation $D = M/R$ (DeGerlia compactness), the governing parameter is the ratio of mass-over-radius terms, $D_1/D_2 = M_1R_2/M_2R_1$, with the Schwarzschild collapse condition expressed as the universal threshold $D_{\text{crit}} = c^2/2G = 6.73295 \times 10^{26}$ kg/m. Time dilation then follows from the same form in every case, $\sqrt{1-k}$, where k is the relevant compactness-derived geometric ratio. Across static mass, static radius, constant density, constant inertia, and fully unconstrained mass-radius comparisons, the compactness-ratio calculation returns the Schwarzschild time-dilation result exactly. This establishes that the relative rate of time between systems is not simply a mass influenced phenomenon largely attributed to proximity to massive objects or degrees ofrelative space-time curvature, but a direct consequence of comparative mass-radius geometry. The conclusions makes explicit a simple relation clearly contained within general relativity: given two systems' masses and mean radii, their relative time dilation is determined by their respective geometric distribution of that mass.

$$\mathrm{GTD} = \sqrt{1-k} \quad \text{where} \quad k = \frac{D_1}{D_{\text{crit}}} = \frac{M_1}{R_1 D_{\text{crit}}} = \frac{r_s}{R} \quad \text{and} \quad D_{\text{crit}} = 6.73295 \times 10^{26} \text{ kg/m}.$$

Keywords: time dilation, general relativity, special relativity, gravitational time dilation, Lorentz time dilation, DeGerlia compactness, inertial density, moment of inertia, Schwarzschild radius

1 Introduction

The Principle of Relativity acknowledges that things can appear differently from different frames of reference. General relativity is certainly no exception. In fact, it is the rule in regard to relativistic behavior. This paper demonstrates that general and special relativity are always comparative models, and that the very structure of general relativity reveals much about this principle that is obscured behind the conventional, sorry, behind the Schwarzschild solution, for lack of a better word. We demonstrate herein a direct relationship between time dilation between two systems is the natural model for evaluating things under general relativity. And in fact the Schwarzschild metric that we currently use for gravitational collapse is just a two system comparison in disguise. To describe it succinctly, The Schwarzschild solution is simply two systems. One being the system you're evaluating, and the other being a system of identical mass at its own Schwarzschild radius. This calculation is understood to produce the relative pace of time between the system being evaluated and what is referred to as "flat space-time" (essentially zero time dilation relative to the system). We show that time, or rather, the pace of time for a system, is emergent of that system's relaive spatial distribution of mass. That spatial distribution is measured axially in units of mass over mean radius, refered to herein as DeGerlia Compactness. Further, we demonstrate that not only is this formula predictive of static mass comparisons such as the short shield condition. It is also universally predictive of other constrained systems we explore herein as well as a universal comparison of two systems of any scale or compactness. Finally begin to introduce a few important related principles that we will expand upon in future papers.

General relativity is a model for observing our universe that is fundamentally comparative: no quantity has physical mean-

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ing without a comparator. The pace of time is no exception: the time dilation between two systems is a a product of the systems respective geometric mass distribution. Directly from the gravitional time dilation formula one can derive the underlying two-system logic and ultimately GTD is simply a comparison between a system a systems pace of time relative to the pace of time of an arbitrarily selected benchmark system, which in this case is the same system at its Schwarzschild radius. The formula allows you the freedom to only insert one mass and one radius, and the second is derived directly from the first. This has the effect of hiding the true nature of relativity and that is that it is, above all, comparative. But sends a signal that one need not compare two systems to understand time, and that is flawed thinking; it introduces the mathematical fallacy of an arbitrarily selected comparator, when there is, in fact, no privilege perspective from which to understand time. In this paper we evaluate the physical property responsible for relativistic time dilation and derive the underlying formulation that quantifies the relative time dilation between two systems.

The relative flow of time between two systems is dictated by their relative compactness. The standard formula for compactness, described under general relativity, expresses this as a dimensionless parameter. However, in this paper we utilize a principle developed by the author, DeGerlia Compactness, which is an abstracted compactness measure explained herein, that retains the proper unit of this property and that is mass over radius. Thus DeGerlia compactness is defined as a system's mass over mean radius $D = m/\bar{r}$[1]. We do a comparative analysis and tabulate relative time dilation across a variety of systems from classical to relativistic regimes, and show that the same compactness ratio $k = D_1/D_2 = M_1R_2/(M_2R_1)$ returns the Schwarzschild time-dilation ratio exactly across every tested constraint class, including the fully general case. This establishes that the relative rate of time between systems is not an ad hoc relativistic output requiring separate formulas, but a direct consequence of comparative mass-radius geometry. The result makes explicit a simple relation already contained in general relativity: given two systems' masses and mean radii, their relative time dilation is determined by their respective geometric distribution of mass alone.

2 Symbol Glossary

Symbol	Definition / value
GTD	$\sqrt{1-k} = d\tau/dt$
DTD	$\sqrt{k}$ (DeGerlia time dilation)
k	$v^2/c^2 = r_s/R = D/D_{\text{crit}}$
D	$M/\bar{R}_{\text{axis}}$ (DeGerlia compactness)
D_{crit}	$c^2/2G = 6.73295 \times 10^{26}$ kg/m
r_s	$2GM/c^2$ (Schwarzschild radius)
P	I/V (inertial density) $= D \cdot 3/10\pi$
θ	$\arcsin\sqrt{k}$; GTD $= \cos\theta$, DTD $= \sin\theta$
c	299,792,458 m/s (exact)
G	6.67430×10^{-11} m^{3}kg^{-1}s^{-2}

3 General Relativity and Compactness

3.1 Compactness

Compactness is the dimensionless ratio of a system's mass to its radius, scaled by G/c^2. The most common forms are $C = GM/(Rc^2)$ and $r_s/R = 2GM/(Rc^2) = 2C$, where $r_s = 2GM/c^2$ is the Schwarzschild radius. Both measure how close a system sits to its Schwarzschild threshold: $r_s/R = 1$ marks the gravitational-collapse condition.

3.2 Time Dilation

3.3 Structure of the GTD Equation as a Two-System Comparison

4 The DeGerlia Time Dilation Framework

An analogous framework that is directly derived from general relativity and has a direct relationship to gravitational time dilation.

4.1 The Scale Factor k

The following identity relates the time-dilation ratio of two uniform spherical systems to their mass and radius.*

$$k = \frac{D_1}{D_2} = \frac{M_1R_2}{M_2R_1} \tag{1}$$

where I_1, I_2 are the moments of inertia, R_1, R_2 are the radii, M_1, M_2 are the masses, and $D_i = M_i/R_i$ is the DeGerlia Compactness of system i. The three forms in Equation 1 are algebraically identical; $(I_1/I_2)(R_2/R_1)^3$ exposes the inertia-radius structure, D_1/D_2 exposes the compactness-ratio structure, and $M_1R_2/(M_2R_1)$ is the fully expanded form.

A system's pace of time has meaning only relative to another pace of time. The scale factor k governs the relative time-dilation between two uniform spherical systems. The standard gravitational time-dilation calculation takes the Schwarzschild form $\sqrt{1-k}$, where $k = D/D_{crit} = r_s/R$. The compactness ratio $k = D_1/D_2$ and the Schwarzschild form give the same value.

Via the Schwarzschild radius, the same scale factor is:

$$k = \frac{D}{D_{crit}} = \frac{r_s}{R} \tag{2}$$

where $D = M/R$ is the system's DeGerlia Compactness and $r_s = 2GM/c^2$ is its Schwarzschild radius. This is the k that appears in the standard $\sqrt{1-k}$ for GTD.

Time dilation is always:

$$\frac{d\tau}{dt} = \sqrt{1-k} \tag{3}$$

where $d\tau/dt$ is the time-dilation factor between the two systems and k is the DeGerlia compactness ratio for the case being computed. The specific forms are:

- $k = D_1/D_2$ (Equation 1) — via inertial density ratio
- $k = D/D_{crit} = r_s/R$ (Equation 2) — via Schwarzschild radius

* This identity has been previously referred to as the "space-time equivalence"[2].

Same k symbol across forms: a given numerical k produces the same time-scale-factor result regardless of which form it appears in.

4.2 DeGerlia Compactness D

Inertial density P[1] is the moment of inertia of a system divided by its volume:

$$P = I/V \tag{4}$$

Inertia, in a linear context, is simply mass. Inertial density, in this context, reduces directly to mass density $\rho = M/V$. From this perspective, inertial density is simply the rotational analog to mass density, not unlike the relationship between f equals m a and torque equals angular acceleration times m. More succinctly stated: mass density is simply the linear interpretation of inertial density.

Substituting $I = \frac{2}{5}MR^2$ and $V = \frac{4}{3}\pi R^3$, with R the radius, this reduces to:

$$P = \frac{M}{R} \times \frac{3}{10\pi} \tag{5}$$

Just as ordinary density ($\rho = M/V$) is mass distributed over volume, inertial density is rotational inertia distributed over volume. As stated in the prior work: *mass and density are to linear motion as moment of inertia and inertial density are to rotational motion.*

$D = M/R$ is a simplified analog to inertial density, related to P by the geometric factor $3/10\pi$, and represents the mass over radius ratio for a system. Given that this quantity is almost always used in a ratio that would cancel out all geometric vectors and constants, for ease of use, this metric D for inertial density is typically used over the geometrically accurate inertial density, P.

4.3 DeGerlia Threshold D_{crit}

The Schwarzschild condition can be expressed as a static universal constant threshold of DeGerlia compactness D. There are several advantages to this approach, not the least of which is that you don't have to calculate the threshold for every mass. And, it's much easier to calculate D than P. Inertial density is an intuitive metric that's easier to work with.

Setting $R = r_s$ (the Schwarzschild condition)[3] and solving yields the **DeGerlia Threshold**: the universal constant of DeGerlia compactness at which the escape velocity equals the speed of light:

$$D_{crit} = \frac{c^2}{2G} = (6.73295 \pm 0.00015) \times 10^{26}\ \text{kg/m} \tag{6}$$

Note this value is exact; its precision is limited by $G = (6.67430 \pm 0.00015) \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$[4].

4.4 DeGerlia Time Dilation

DeGerlia Time Dilation is the geometric complement of gravitational time dilation:

$$\text{DTD} = \sqrt{k} \tag{7}$$

where k is the scale factor of Equation 1. Where GTD $= \sqrt{1-k}$ is the surviving proper-time rate $d\tau/dt$, DTD $= \sqrt{k}$ is its complement.

4.5 Converting Between GTD and DTD

GTD and DTD are bound by the Pythagorean identity:

$$\text{GTD}^2 + \text{DTD}^2 = (1-k) + k = 1. \tag{8}$$

4.6 Properties of DTD

5 Relative Time Dilation Examples

Example parameter sets match the columns of the pair comparison tables in Appendix A.

5.1 Static Mass ($M_1 = M_2$)

When the two systems share the same mass, the mass terms cancel in k:

$$k = \frac{M_1 R_2}{M_2 R_1} = \frac{R_2}{R_1}$$

The compactness ratio reduces to a ratio of radii. Equivalently in terms of DeGerlia compactness $D = M/R$, this is $D_1/D_2 = R_2/R_1$; each system's normalized DeGerlia compactness against the DeGerlia Threshold yields $D/D_{crit} = r_s/R$, recovering the Schwarzschild gravitational time dilation parameter.

Example (Static Mass): System 2 is the system being measured, at $\frac{4}{3}$ its Schwarzschild radius; System 1 is a same-mass reference set to its Schwarzschild radius ($R_1 = r_s$).

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000×10^{30}	1.00000×10^{30}
R (m)	r_s	$\frac{4}{3} r_s$

Calculate DTD for each System:

$$\text{DTD} = \sqrt{K} \quad \text{where} \quad K = \frac{M}{R\,D_{\text{crit}}}$$

$$K_1 = \frac{M_1}{R_1\,D_{\text{crit}}} = \frac{r_s}{R_1} = \frac{r_s}{r_s} = 1$$

$$\text{DTD}_1 = \sqrt{1} = 1$$

$$K_2 = \frac{M_2}{R_2\,D_{\text{crit}}} = \frac{r_s}{R_2} = \frac{r_s}{\frac{4}{3} r_s} = \frac{3}{4}$$

$$\text{DTD}_2 = \sqrt{\tfrac{3}{4}} = 8.66025 \times 10^{-1}$$

$$\frac{\text{DTD}_1}{\text{DTD}_2} = \frac{1}{8.66025 \times 10^{-1}} = 1.15470$$

Calculate GTD ratio from DTD:

$$\text{GTD}_1/\text{GTD}_2 = \sqrt{1-\text{DTD}_1^2} \Big/ \sqrt{1-\text{DTD}_2^2}$$
$$= \sqrt{1-(1)^2} \Big/ \sqrt{1-(8.66025 \times 10^{-1})^2}$$

$$= \sqrt{1-1} \Big/ \sqrt{1-\tfrac{3}{4}} = \sqrt{0} \Big/ \sqrt{\tfrac{1}{4}} = 0 \big/ \tfrac{1}{2} = 0$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:
Schwarzschild radii:

$$r_{s1} = r_{s2} = \frac{2GM_1}{c^2}$$

Calculate K values from the r_s/R ratio:

$$K_1 = \frac{r_{s1}}{R_1} \qquad = \frac{r_{s1}}{r_{s1}} \qquad = 1$$

$$K_2 = \frac{r_{s2}}{R_2} \qquad = \frac{r_{s2}}{\frac{4}{3}\, r_{s2}} \qquad = \frac{3}{4}$$

Calculate GTD via Schwarzschild radius ratio:

$$\mathrm{GTD}_1 = \sqrt{1-K_1} = \sqrt{1-1} = \sqrt{0} = 0$$

$$\mathrm{GTD}_2 = \sqrt{1-K_2} = \sqrt{1-\tfrac{3}{4}} = \sqrt{\tfrac{1}{4}} = \tfrac{1}{2}$$

$$\mathrm{GTD}_1/\mathrm{GTD}_2 = 0 \big/ \tfrac{1}{2} = 0$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\mathrm{GTD}_1/\mathrm{GTD}_2 = 0 \quad (\text{via } r_s)$$
$$\mathrm{GTD}_1/\mathrm{GTD}_2 = 0 \quad (\text{via } D_{\text{crit}})$$

$$\text{difference} = 0 \quad (\text{both routes give the same value})$$

All values above appear in column $M = M$ of Table 1 in Appendix A.

5.2 Example (Static Mass, Table Case 1)

Example (Static Mass, Table Case 1): System 2 is the system being measured; System 1 is a same-mass reference system set to its Schwarzschild radius ($R_1 = r_s$).

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	2.00000×10^{30}	2.00000×10^{30}
R (m)	r_s	7.00000×10^{8}

Calculate DTD for each System:

$$\mathrm{DTD} = \sqrt{K} \quad \text{where} \quad K = \frac{M}{R\,D_{\text{crit}}}$$

$$K_1 = \frac{M_1}{R_1\,D_{\text{crit}}} = \frac{r_s}{R_1} = \frac{r_s}{r_s} = 1$$

$$\mathrm{DTD}_1 = \sqrt{1} = 1$$

$$K_2 = \frac{2.00000 \times 10^{30}\,\mathrm{kg}}{(7.00000 \times 10^{8}\,\mathrm{m})(6.73295 \times 10^{26}\,\mathrm{kg/m})} = 4.24352 \times 10^{-6}$$

$$\mathrm{DTD}_2 = \sqrt{4.24352 \times 10^{-6}} = 2.05998 \times 10^{-3}$$

$$\frac{\mathrm{DTD}_1}{\mathrm{DTD}_2} = \frac{1}{2.05998 \times 10^{-3}} = 4.85441 \times 10^{2}$$

Calculate GTD ratio from DTD:

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = \frac{\sqrt{1-\mathrm{DTD}_1^2}}{\sqrt{1-\mathrm{DTD}_2^2}} = \frac{\sqrt{1-(1)^2}}{\sqrt{1-(2.05998 \times 10^{-3})^2}}$$
$$= \frac{\sqrt{1-1}}{\sqrt{1-4.24352 \times 10^{-6}}} = \frac{\sqrt{0}}{\sqrt{9.9999575648 \times 10^{-1}}}$$
$$= \frac{0}{9.9999787824 \times 10^{-1}} = 0$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:
Schwarzschild radii:

$$r_{s1} = r_{s2} = \frac{2GM_1}{c^2} = 2.97046 \times 10^{3}\,\mathrm{m}$$

Calculate K values from the r_s/R ratio:

$$K_1 = \frac{r_{s1}}{R_1} \qquad = \frac{r_{s1}}{r_{s1}} \qquad = 1$$

$$K_2 = \frac{r_{s2}}{R_2} \qquad = \frac{2.97046 \times 10^{3}\,\mathrm{m}}{7.00000 \times 10^{8}\,\mathrm{m}} \qquad = 4.24352 \times 10^{-6}$$

Calculate GTD via Schwarzschild radius ratio:

$$\mathrm{GTD}_1 = \sqrt{1-K_1} = \sqrt{1-1} = \sqrt{0} = 0$$

$$\mathrm{GTD}_2 = \sqrt{1-K_2} = \sqrt{1-4.24352 \times 10^{-6}}$$
$$= \sqrt{9.9999575648 \times 10^{-1}}$$
$$= 9.9999787824 \times 10^{-1}$$

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = \frac{0}{9.9999787824 \times 10^{-1}} = 0$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{\mathrm{GTD}_{S1}}{\mathrm{GTD}_{S2}} = 0 \quad (\text{via } r_s)$$
$$\frac{\mathrm{GTD}_{D1}}{\mathrm{GTD}_{D2}} = 0 \quad (\text{via } D_{\text{crit}})$$

$$\text{difference} = 0 \quad (\text{both routes give exactly zero})$$

All values above appear in column $M = M$ of Table 1 in Appendix A.

5.3 Example - Static Mass , Table 2 reference M=M Case

When the two systems share the same mass, the mass terms cancel in k_d:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \frac{R_2}{R_1}$$

The compactness ratio reduces to a ratio of radii. Equivalently in terms of DeGerlia compactness $D = M/R$, this is $D_1/D_2 = R_2/R_1$; each system's normalized DeGerlia compactness against the DeGerlia Threshold yields $D/D_{crit} = r_s/R$, recovering the Schwarzschild gravitational time dilation parameter.

This case is the standard gravitational time-dilation calculation; the $M_1R_2/(M_2R_1)$ ratio reduces to the familiar Schwarzschild factor here. Both systems are written out and computed independently below — the same Schwarzschild calculation, with the second system (the same mass at its Schwarzschild radius) made visible instead of folded into r_s.

Example (Static Mass, Table Case 1): System 1 is the system being measured; System 2 is a same-mass reference system at a different radius.

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	2.00000×10^{30}	2.00000×10^{30}
R (m)	3.00000×10^{3}	7.00000×10^{8}

Calculate DTD for each System:

$$\mathrm{DTD} = \sqrt{K} \quad \text{where} \quad K = \frac{M}{R\,D_{\mathrm{crit}}}$$

$$K_1 = \frac{2.00000 \times 10^{30}\,\mathrm{kg}}{(3.00000 \times 10^{3}\,\mathrm{m})(6.73295 \times 10^{26}\,\mathrm{kg/m})} = 9.9015538 \times 10^{-1}$$

$$\mathrm{DTD}_1 = \sqrt{9.9015538 \times 10^{-1}} = 9.9506552 \times 10^{-1}$$

$$K_2 = \frac{2.00000 \times 10^{30}\,\mathrm{kg}}{(7.00000 \times 10^{8}\,\mathrm{m})(6.73295 \times 10^{26}\,\mathrm{kg/m})} = 4.24352 \times 10^{-6}$$

$$\mathrm{DTD}_2 = \sqrt{4.24352 \times 10^{-6}} = 2.05998 \times 10^{-3}$$

$$\frac{\mathrm{DTD}_1}{\mathrm{DTD}_2} = \frac{9.9506552 \times 10^{-1}}{2.05998 \times 10^{-3}} = 4.83046 \times 10^{2}$$

Calculate GTD ratio from DTD:

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = \frac{\sqrt{1-\mathrm{DTD}_1^2}}{\sqrt{1-\mathrm{DTD}_2^2}} = \frac{\sqrt{1-(9.9506552 \times 10^{-1})^2}}{\sqrt{1-(2.05998 \times 10^{-3})^2}}$$
$$= \frac{\sqrt{1-9.9015539 \times 10^{-1}}}{\sqrt{1-4.24352 \times 10^{-6}}}$$
$$= \frac{\sqrt{9.84461 \times 10^{-3}}}{\sqrt{9.999957565 \times 10^{-1}}}$$
$$= \frac{9.92200 \times 10^{-2}}{9.999978782 \times 10^{-1}} = 9.9220223 \times 10^{-2}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:

Schwarzschild radii:

$$r_{s1} = r_{s2} = \frac{2GM_1}{c^2} = 2.97046 \times 10^{3}\,\mathrm{m}$$

Calculate K values from the r_s/R ratio:

$$K_1 = \frac{r_{s1}}{R_1} = \frac{2.97046 \times 10^{3}\,\mathrm{m}}{3.00000 \times 10^{3}\,\mathrm{m}} = 9.9015333 \times 10^{-1}$$

$$K_2 = \frac{r_{s2}}{R_2} = \frac{2.97046 \times 10^{3}\,\mathrm{m}}{7.00000 \times 10^{8}\,\mathrm{m}} = 4.24352 \times 10^{-6}$$

Calculate GTD via Schwarzschild radius ratio:

$$\mathrm{GTD}_1 = \sqrt{1-K_1} = \sqrt{1-9.9015333 \times 10^{-1}}$$
$$= \sqrt{9.84667 \times 10^{-3}} = 9.92303 \times 10^{-2}$$
$$= 1 - 9.007697 \times 10^{-1}$$

$$\mathrm{GTD}_2 = \sqrt{1-K_2} = \sqrt{1-4.24352 \times 10^{-6}}$$
$$= \sqrt{9.9999575648 \times 10^{-1}}$$
$$= 9.9999787824 \times 10^{-1} = 1 - 2.12180 \times 10^{-6}$$

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = \frac{9.92303 \times 10^{-2}}{9.9999787824 \times 10^{-1}}$$
$$= 9.92305 \times 10^{-2}$$
$$= 1 - 9.007695 \times 10^{-1}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\frac{\mathrm{GTD}_{S1}}{\mathrm{GTD}_{S2}} = 9.92305 \times 10^{-2} \quad (\text{via } r_s)$$
$$\frac{\mathrm{GTD}_{D1}}{\mathrm{GTD}_{D2}} = 9.92203 \times 10^{-2} \quad (\text{via } D_{\mathrm{crit}})$$

difference = 0.000002

(below the margin of error imparted by G, 0.00015, the gravitational constant's limit)

All values above appear in column $M = M$ of Table 1 in Appendix A.

5.4 Static Radius (Linear) ($R_1 = R_2$)

When the two systems share the same radius, the radius terms cancel in k_d:

$$k_d = \frac{M_1R_2}{M_2R_1} = \frac{M_1}{M_2}$$

The DeGerlia compactness ratio reduces to a ratio of masses. The Lorentz factor has the same form as Equation 3, with $k = v^2/c^2$. Since velocity is distance per unit time, v^2/c^2 is the square of a ratio of two lengths, with the shared time unit cancelling. Gravitational time dilation confirms this directly: $v^2/c^2 = r_s/R$ via the escape velocity identity, so the velocity-based k of special relativity and the radius-based k of general relativity are the same length ratio.

Example (Static Radius):

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000×10^{24}	1.00000×10^{30}
R (m)	1.00000×10^{8}	1.00000×10^{8}

Calculate DTD for each System:

$$\mathrm{DTD} = \sqrt{K} \quad \text{where} \quad K = M/(R\,D_{\mathrm{crit}})$$

$$K_1 = \frac{1.00000 \times 10^{24}\,\mathrm{kg}}{(1.00000 \times 10^{8}\,\mathrm{m})(6.73295 \times 10^{26}\,\mathrm{kg/m})} = 1.48523 \times 10^{-11}$$

$$\mathrm{DTD}_1 = \sqrt{1.48523 \times 10^{-11}} = 3.85387 \times 10^{-6}$$

$$K_2 = \frac{1.00000 \times 10^{30}\,\mathrm{kg}}{(1.00000 \times 10^{8}\,\mathrm{m})(6.73295 \times 10^{26}\,\mathrm{kg/m})} = 1.48523 \times 10^{-5}$$

$$\mathrm{DTD}_2 = \sqrt{1.48523 \times 10^{-5}} = 3.85387 \times 10^{-3}$$

$$\frac{\mathrm{DTD}_1}{\mathrm{DTD}_2} = \frac{3.85387 \times 10^{-6}}{3.85387 \times 10^{-3}} = 1.00000 \times 10^{-3}$$

Calculate GTD ratio from DTD:

$$\begin{aligned}
\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} &= \frac{\sqrt{1-\mathrm{DTD}_1^2}}{\sqrt{1-\mathrm{DTD}_2^2}} \\
&= \frac{\sqrt{1-(3.85387 \times 10^{-6})^2}}{\sqrt{1-(3.85387 \times 10^{-3})^2}} \\
&= \frac{\sqrt{1-1.48523 \times 10^{-11}}}{\sqrt{1-1.48523 \times 10^{-5}}} \\
&= \frac{\sqrt{9.999999999851477 \times 10^{-1}}}{\sqrt{9.999851477 \times 10^{-1}}} \\
&= \frac{9.9999999999257383 \times 10^{-1}}{9.9999257381 \times 10^{-1}} \\
&= 1.00001
\end{aligned}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:

Schwarzschild radii:

$$\begin{aligned}
r_{s1} &= \frac{2GM_1}{c^2} \\
&= \frac{2(6.67430 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(1.00000 \times 10^{24}\,\mathrm{kg})}{(2.99792 \times 10^{8}\,\mathrm{m\,s^{-1}})^2} \\
&= 1.48523 \times 10^{-3}\,\mathrm{m}
\end{aligned}$$

$$\begin{aligned}
r_{s2} &= \frac{2GM_2}{c^2} \\
&= \frac{2(6.67430 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(1.00000 \times 10^{30}\,\mathrm{kg})}{(2.99792 \times 10^{8}\,\mathrm{m\,s^{-1}})^2} \\
&= 1.48523 \times 10^{3}\,\mathrm{m}
\end{aligned}$$

Calculate K values from the r_s/R ratio:

$$K_1 = \frac{r_{s1}}{R_1} = \frac{1.48523 \times 10^{-3}\,\mathrm{m}}{1.00000 \times 10^{8}\,\mathrm{m}} = 1.48523 \times 10^{-11}$$

$$K_2 = \frac{r_{s2}}{R_2} = \frac{1.48523 \times 10^{3}\,\mathrm{m}}{1.00000 \times 10^{8}\,\mathrm{m}} = 1.48523 \times 10^{-5}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
\mathrm{GTD}_1 &= \sqrt{1-K_1} = \sqrt{1-1.48523 \times 10^{-11}} \\
&= \sqrt{9.999999999851477 \times 10^{-1}} \\
&= 9.9999999999257384 \times 10^{-1} \\
&= 1 - 7.42616 \times 10^{-12}
\end{aligned}$$

$$\begin{aligned}
\mathrm{GTD}_2 &= \sqrt{1-K_2} = \sqrt{1-1.48523 \times 10^{-5}} \\
&= \sqrt{9.999851477 \times 10^{-1}} \\
&= 9.9999257381 \times 10^{-1} \\
&= 1 - 7.42619 \times 10^{-6}
\end{aligned}$$

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = \frac{9.9999999999257384 \times 10^{-1}}{9.9999257381 \times 10^{-1}} = 1.00001$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\mathrm{GTD}_1/\mathrm{GTD}_2 = 1.00001 \quad (\text{via } r_s)$$
$$\mathrm{GTD}_1/\mathrm{GTD}_2 = 1.00001 \quad (\text{via } D_{\mathrm{crit}})$$

difference = 0

(below the margin of error imparted by G, 0.00015, the gravitational constant's limit)

This case is tabulated in Table 2, column $\rho = \rho$ of Appendix A.

5.5 Constant Density ($\rho_1 = \rho_2$)

When the two systems share the same density, $M \propto R^3$, so $M_1/M_2 = (R_1/R_2)^3$. Substituting into k_d:

$$k_d = \frac{M_1 R_2}{M_2 R_1} = \left(\frac{R_1}{R_2}\right)^3 \cdot \frac{R_2}{R_1} = \left(\frac{R_1}{R_2}\right)^2$$

The DeGerlia compactness ratio reduces to the square of the radius ratio; the linear scale factor follows as $\sqrt{k_d} = R_1/R_2$, recovering the isometric equivalence.

Example (Constant Density):

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000×10^{30}	1.00000×10^{27}
R (m)	1.00000×10^{8}	1.00000×10^{7}

Calculate DTD for each System:

$$\mathrm{DTD} = \sqrt{K} \quad \text{where} \quad K = M/(R\,D_{\mathrm{crit}})$$

$$K_1 = \frac{1.00000 \times 10^{30}\,\mathrm{kg}}{(1.00000 \times 10^{8}\,\mathrm{m})(6.73295 \times 10^{26}\,\mathrm{kg/m})} = 1.48523 \times 10^{-5}$$

$$\mathrm{DTD}_1 = \sqrt{1.48523 \times 10^{-5}} = 3.85387 \times 10^{-3}$$

$$K_2 = \frac{1.00000 \times 10^{27}\,\mathrm{kg}}{(1.00000 \times 10^{7}\,\mathrm{m})(6.73295 \times 10^{26}\,\mathrm{kg/m})} = 1.48523 \times 10^{-7}$$

$$\mathrm{DTD}_2 = \sqrt{1.48523 \times 10^{-7}} = 3.85387 \times 10^{-4}$$

$$\frac{\mathrm{DTD}_1}{\mathrm{DTD}_2} = \frac{3.85387 \times 10^{-3}}{3.85387 \times 10^{-4}} = 1.00000 \times 10^{1}$$

Calculate GTD ratio from DTD:

$$\begin{aligned}
\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} &= \frac{\sqrt{1-\mathrm{DTD}_1^2}}{\sqrt{1-\mathrm{DTD}_2^2}} \\
&= \frac{\sqrt{1-(3.85387 \times 10^{-3})^2}}{\sqrt{1-(3.85387 \times 10^{-4})^2}} \\
&= \frac{\sqrt{1-1.48523 \times 10^{-5}}}{\sqrt{1-1.48523 \times 10^{-7}}} \\
&= \frac{\sqrt{9.999851477 \times 10^{-1}}}{\sqrt{9.99999851477 \times 10^{-1}}} \\
&= \frac{9.9999257381 \times 10^{-1}}{9.999999257383 \times 10^{-1}} \\
&= 9.9999264807 \times 10^{-1} \\
&= 1 - 7.35193 \times 10^{-6}
\end{aligned}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:

Schwarzschild radii:

$$\begin{aligned}
r_{s1} &= \frac{2GM_1}{c^2} \\
&= \frac{2(6.67430 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(1.00000 \times 10^{30}\,\mathrm{kg})}{(2.99792 \times 10^{8}\,\mathrm{m\,s^{-1}})^2} \\
&= 1.48523 \times 10^{3}\,\mathrm{m}
\end{aligned}$$

$$\begin{aligned}
r_{s2} &= \frac{2GM_2}{c^2} \\
&= \frac{2(6.67430 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(1.00000 \times 10^{27}\,\mathrm{kg})}{(2.99792 \times 10^{8}\,\mathrm{m\,s^{-1}})^2} \\
&= 1.48523\,\mathrm{m}
\end{aligned}$$

Calculate K values from the r_s/R ratio:

$$\begin{aligned}
K_1 &= \frac{r_{s1}}{R_1} &&= \frac{1.48523 \times 10^{3}\,\mathrm{m}}{1.00000 \times 10^{8}\,\mathrm{m}} &&= 1.48523 \times 10^{-5} \\
K_2 &= \frac{r_{s2}}{R_2} &&= \frac{1.48523\,\mathrm{m}}{1.00000 \times 10^{7}\,\mathrm{m}} &&= 1.48523 \times 10^{-7}
\end{aligned}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
\mathrm{GTD}_1 &= \sqrt{1-K_1} = \sqrt{1-1.48523 \times 10^{-5}} \\
&= \sqrt{9.999851477 \times 10^{-1}} \\
&= 9.9999257381 \times 10^{-1} \\
&= 1 - 7.42619 \times 10^{-6}
\end{aligned}$$

$$\begin{aligned}
\mathrm{GTD}_2 &= \sqrt{1-K_2} = \sqrt{1-1.48523 \times 10^{-7}} \\
&= \sqrt{9.99999851477 \times 10^{-1}} \\
&= 9.999999257384 \times 10^{-1} \\
&= 1 - 7.42616 \times 10^{-8}
\end{aligned}$$

$$\begin{aligned}
\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} &= \frac{9.9999257381 \times 10^{-1}}{9.999999257384 \times 10^{-1}} \\
&= 9.9999264807 \times 10^{-1} \\
&= 1 - 7.35193 \times 10^{-6}
\end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\mathrm{GTD}_1/\mathrm{GTD}_2 = 1 - 7.35193 \times 10^{-6} \quad (\text{via } r_s)$$
$$\mathrm{GTD}_1/\mathrm{GTD}_2 = 1 - 7.35193 \times 10^{-6} \quad (\text{via } D_{\mathrm{crit}})$$

difference = 0

(below the margin of error imparted by G, 0.00015, the gravitational constant's limit)

This case is tabulated in Table 2, column $\rho = \rho$ of Appendix A.

5.6 Constant Inertia ($I_1 = I_2$)

When the two systems share the same moment of inertia, $M_1R_1^2 = M_2R_2^2$, so $M_1/M_2 = (R_2/R_1)^2$. Substituting into k_d:

$$k_d = \frac{M_1R_2}{M_2R_1} = \left(\frac{R_2}{R_1}\right)^2 \cdot \frac{R_2}{R_1} = \left(\frac{R_2}{R_1}\right)^3$$

The DeGerlia compactness ratio reduces to the cube of the radius ratio.

Example (Constant Inertia):

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	1.00000×10^{32}	1.00000×10^{30}
R (m)	1.00000×10^{7}	1.00000×10^{8}

Calculate DTD for each System:

$$\mathrm{DTD} = \sqrt{K} \quad \text{where} \quad K = M/(RD_{\mathrm{crit}})$$

$$K_1 = \frac{1.00000 \times 10^{32}\,\mathrm{kg}}{(1.00000 \times 10^{7}\,\mathrm{m})(6.73295 \times 10^{26}\,\mathrm{kg/m})} = 1.48523 \times 10^{-2}$$

$$\mathrm{DTD}_1 = \sqrt{1.48523 \times 10^{-2}} = 1.21870 \times 10^{-1}$$

$$K_2 = \frac{1.00000 \times 10^{30}\,\mathrm{kg}}{(1.00000 \times 10^{8}\,\mathrm{m})(6.73295 \times 10^{26}\,\mathrm{kg/m})} = 1.48523 \times 10^{-5}$$

$$\mathrm{DTD}_2 = \sqrt{1.48523 \times 10^{-5}} = 3.85387 \times 10^{-3}$$

$$\frac{\mathrm{DTD}_1}{\mathrm{DTD}_2} = \frac{1.218\,70 \times 10^{-1}}{3.853\,87 \times 10^{-3}} = 3.162\,28 \times 10^{1}$$

Calculate GTD ratio from DTD:

$$\begin{aligned}
\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} &= \frac{\sqrt{1-\mathrm{DTD}_1^2}}{\sqrt{1-\mathrm{DTD}_2^2}} \\
&= \frac{\sqrt{1-(1.218\,70\times10^{-1})^2}}{\sqrt{1-(3.853\,87\times10^{-3})^2}} \\
&= \frac{\sqrt{9.851\,477\times10^{-1}}}{\sqrt{9.999\,851\,477\times10^{-1}}} \\
&= \frac{9.925\,460\,5\times10^{-1}}{9.999\,925\,738\,1\times10^{-1}} \\
&= 9.925\,534\,2\times10^{-1} \\
&= 1-7.446\,58\times10^{-3}
\end{aligned}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:

Schwarzschild radii:

$$\begin{aligned}
r_{s1} &= \frac{2GM_1}{c^2} \\
&= \frac{2(6.674\,30\times10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(1.000\,00\times10^{32}\,\mathrm{kg})}{(2.997\,92\times10^{8}\,\mathrm{m\,s^{-1}})^2} \\
&= 1.485\,23\times10^{5}\,\mathrm{m}
\end{aligned}$$

$$\begin{aligned}
r_{s2} &= \frac{2GM_2}{c^2} \\
&= \frac{2(6.674\,30\times10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(1.000\,00\times10^{30}\,\mathrm{kg})}{(2.997\,92\times10^{8}\,\mathrm{m\,s^{-1}})^2} \\
&= 1.485\,23\times10^{3}\,\mathrm{m}
\end{aligned}$$

Calculate K values from the r_s/R ratio:

$$\begin{aligned}
K_1 &= \frac{r_{s1}}{R_1} &&= \frac{1.485\,23\times10^{5}\,\mathrm{m}}{1.000\,00\times10^{7}\,\mathrm{m}} &&= 1.485\,23\times10^{-2} \\
K_2 &= \frac{r_{s2}}{R_2} &&= \frac{1.485\,23\times10^{3}\,\mathrm{m}}{1.000\,00\times10^{8}\,\mathrm{m}} &&= 1.485\,23\times10^{-5}
\end{aligned}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
\mathrm{GTD}_1 &= \sqrt{1-K_1} = \sqrt{1-1.485\,23\times10^{-2}} \\
&= \sqrt{9.851\,477\times10^{-1}} \\
&= 9.925\,460\,6\times10^{-1} \\
&= 1-7.453\,94\times10^{-3}
\end{aligned}$$

$$\begin{aligned}
\mathrm{GTD}_2 &= \sqrt{1-K_2} = \sqrt{1-1.485\,23\times10^{-5}} \\
&= \sqrt{9.999\,851\,477\times10^{-1}} \\
&= 9.999\,925\,738\,1\times10^{-1} \\
&= 1-7.426\,19\times10^{-6}
\end{aligned}$$

$$\begin{aligned}
\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} &= \frac{9.925\,460\,6\times10^{-1}}{9.999\,925\,738\,1\times10^{-1}} \\
&= 9.925\,534\,3\times10^{-1} \\
&= 1-7.446\,57\times10^{-3}
\end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\begin{aligned}
\mathrm{GTD}_1/\mathrm{GTD}_2 &= 1-7.446\,57\times10^{-3} \quad (\text{via } r_s) \\
\mathrm{GTD}_1/\mathrm{GTD}_2 &= 1-7.446\,58\times10^{-3} \quad (\text{via } D_{\mathrm{crit}})
\end{aligned}$$

$$\text{difference} = 5.115\,04\times10^{-9}$$

(below the margin of error imparted by G, 0.00015, the gravitational constant's limit)

This case is tabulated in Table 2, column $I = I$ of Appendix A.

5.7 General Case (unconstrained)

When no constraint is applied between the systems, k_d stays in its full form:

$$k_d = \frac{M_1 R_2}{M_2 R_1}$$

Both mass and radius must be specified independently for each system.

Example (General Case):

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	$1.000\,00\times10^{20}$	$1.000\,00\times10^{30}$
R (m)	$2.000\,00\times10^{-7}$	$1.000\,00\times10^{7}$

Calculate DTD for each System:

$$\mathrm{DTD} = \sqrt{K} \quad \text{where} \quad K = M/(R D_{\mathrm{crit}})$$

$$\begin{aligned}
K_1 &= \frac{1.000\,00\times10^{20}\,\mathrm{kg}}{(2.000\,00\times10^{-7}\,\mathrm{m})(6.732\,95\times10^{26}\,\mathrm{kg/m})} \\
&= 7.426\,17\times10^{-1}
\end{aligned}$$

$$\mathrm{DTD}_1 = \sqrt{7.426\,17\times10^{-1}} = 8.617\,52\times10^{-1}$$

$$\begin{aligned}
K_2 &= \frac{1.000\,00\times10^{30}\,\mathrm{kg}}{(1.000\,00\times10^{7}\,\mathrm{m})(6.732\,95\times10^{26}\,\mathrm{kg/m})} \\
&= 1.485\,23\times10^{-4}
\end{aligned}$$

$$\mathrm{DTD}_2 = \sqrt{1.485\,23\times10^{-4}} = 1.218\,70\times10^{-2}$$

$$\frac{\mathrm{DTD}_1}{\mathrm{DTD}_2} = \frac{8.617\,52\times10^{-1}}{1.218\,70\times10^{-2}} = 7.071\,07\times10^{1}$$

Calculate GTD ratio from DTD:

$$\begin{aligned}
\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} &= \frac{\sqrt{1-\mathrm{DTD}_1^2}}{\sqrt{1-\mathrm{DTD}_2^2}} \\
&= \frac{\sqrt{1-(8.617\,52\times10^{-1})^2}}{\sqrt{1-(1.218\,70\times10^{-2})^2}}
\end{aligned}$$

$$= \frac{\sqrt{1 - 7.42617 \times 10^{-1}}}{\sqrt{1 - 1.48523 \times 10^{-4}}}$$
$$= \frac{\sqrt{2.57383 \times 10^{-1}}}{\sqrt{9.99851477 \times 10^{-1}}}$$
$$= \frac{5.07330 \times 10^{-1}}{9.999257356 \times 10^{-1}}$$
$$= 5.07367 \times 10^{-1}$$
$$= 1 - 4.92633 \times 10^{-1}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:
Schwarzschild radii:

$$r_{s1} = \frac{2GM_1}{c^2}$$
$$= \frac{2(6.67430 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(1.00000 \times 10^{20}\,\mathrm{kg})}{(2.99792 \times 10^8\,\mathrm{m\,s^{-1}})^2}$$
$$= 1.48523 \times 10^{-7}\,\mathrm{m}$$

$$r_{s2} = \frac{2GM_2}{c^2}$$
$$= \frac{2(6.67430 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(1.00000 \times 10^{30}\,\mathrm{kg})}{(2.99792 \times 10^8\,\mathrm{m\,s^{-1}})^2}$$
$$= 1.48523 \times 10^{3}\,\mathrm{m}$$

Calculate K values from the r_s/R ratio:

$$K_1 = \frac{r_{s1}}{R_1} = \frac{1.48523 \times 10^{-7}\,\mathrm{m}}{2.00000 \times 10^{-7}\,\mathrm{m}} = 7.42616 \times 10^{-1}$$
$$K_2 = \frac{r_{s2}}{R_2} = \frac{1.48523 \times 10^{3}\,\mathrm{m}}{1.00000 \times 10^{7}\,\mathrm{m}} = 1.48523 \times 10^{-4}$$

Calculate GTD via Schwarzschild radius ratio:

$$\mathrm{GTD}_1 = \sqrt{1 - K_1} = \sqrt{1 - 7.42616 \times 10^{-1}}$$
$$= \sqrt{2.57384 \times 10^{-1}}$$
$$= 5.07330 \times 10^{-1}$$
$$\mathrm{GTD}_2 = \sqrt{1 - K_2} = \sqrt{1 - 1.48523 \times 10^{-4}}$$
$$= \sqrt{9.99851477 \times 10^{-1}}$$
$$= 9.999257356 \times 10^{-1}$$
$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = \frac{5.07330 \times 10^{-1}}{9.999257356 \times 10^{-1}}$$
$$= 5.07368 \times 10^{-1}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\mathrm{GTD}_1/\mathrm{GTD}_2 = 5.07368 \times 10^{-1} \quad (\text{via } r_s)$$
$$\mathrm{GTD}_1/\mathrm{GTD}_2 = 5.07367 \times 10^{-1} \quad (\text{via } D_{\mathrm{crit}})$$

$$\text{difference} = 5.00857 \times 10^{-7}$$

(below the margin of error imparted by G, 0.00015, the gravitational constant's limit)

This case is tabulated in Table 2, column General of Appendix A.

5.8 Classical Static Density Example

Example (Classical):

System properties:

	System 1 (S_1)	System 2 (S_2)
M (kg)	8.00000	1.00000
R (m)	2.00000	1.00000

Calculate DTD for each System:

$$\mathrm{DTD} = \sqrt{K} \quad \text{where} \quad K = M/(RD_{\mathrm{crit}})$$
$$K_1 = \frac{8.00000\,\mathrm{kg}}{(2.00000\,\mathrm{m})(6.73295 \times 10^{26}\,\mathrm{kg/m})}$$
$$= 5.94093 \times 10^{-27}$$
$$\mathrm{DTD}_1 = \sqrt{5.94093 \times 10^{-27}} = 7.70774 \times 10^{-14}$$
$$K_2 = \frac{1.00000\,\mathrm{kg}}{(1.00000\,\mathrm{m})(6.73295 \times 10^{26}\,\mathrm{kg/m})}$$
$$= 1.48523 \times 10^{-27}$$
$$\mathrm{DTD}_2 = \sqrt{1.48523 \times 10^{-27}} = 3.85387 \times 10^{-14}$$
$$\frac{\mathrm{DTD}_1}{\mathrm{DTD}_2} = \frac{7.70774 \times 10^{-14}}{3.85387 \times 10^{-14}} = 2.00000$$

Calculate GTD ratio from DTD:

$$\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} = \frac{\sqrt{1 - \mathrm{DTD}_1^2}}{\sqrt{1 - \mathrm{DTD}_2^2}}$$
$$= \frac{\sqrt{1 - (7.70774 \times 10^{-14})^2}}{\sqrt{1 - (3.85387 \times 10^{-14})^2}}$$
$$= \frac{\sqrt{1 - 5.94093 \times 10^{-27}}}{\sqrt{1 - 1.48523 \times 10^{-27}}}$$
$$= \frac{\sqrt{9.99999999999999999999999999405907 \times 10^{-1}}}{\sqrt{9.99999999999999999999999999851477 \times 10^{-1}}}$$
$$= \frac{9.99999999999999999999999999702953 \times 10^{-1}}{9.99999999999999999999999999257383 \times 10^{-1}}$$
$$= 9.99999999999999999999999999777215 \times 10^{-1}$$
$$= 1 - 2.22785 \times 10^{-27}$$

Verify by Calculating GTD from Schwarzschild Radius Ratio:
Schwarzschild radii:

$$r_{s1} = \frac{2GM_1}{c^2}$$
$$= \frac{2(6.67430 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(8.00000\,\mathrm{kg})}{(2.99792 \times 10^8\,\mathrm{m\,s^{-1}})^2}$$
$$= 1.18819 \times 10^{-26}\,\mathrm{m}$$

$$r_{s2} = \frac{2GM_2}{c^2}$$
$$= \frac{2(6.67430 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}})(1.00000\,\mathrm{kg})}{(2.99792 \times 10^8\,\mathrm{m\,s^{-1}})^2}$$
$$= 1.48523 \times 10^{-27}\,\mathrm{m}$$

Calculate K values from the r_s/R ratio:

$$K_1 = \frac{r_{s1}}{R_1} = \frac{1.18819 \times 10^{-26}\,\mathrm{m}}{2.00000\,\mathrm{m}} = 5.94093 \times 10^{-27}$$
$$K_2 = \frac{r_{s2}}{R_2} = \frac{1.48523 \times 10^{-27}\,\mathrm{m}}{1.00000\,\mathrm{m}} = 1.48523 \times 10^{-27}$$

Calculate GTD via Schwarzschild radius ratio:

$$\begin{aligned}
\mathrm{GTD}_1 &= \sqrt{1-K_1} = \sqrt{1-5.94093 \times 10^{-27}} \\
&= \sqrt{9.99999999999999999999999999405907 \times 10^{-1}} \\
&= 9.99999999999999999999999999702954 \times 10^{-1} \\
&= 1 - 2.97046 \times 10^{-27}
\end{aligned}$$
$$\begin{aligned}
\mathrm{GTD}_2 &= \sqrt{1-K_2} = \sqrt{1-1.48523 \times 10^{-27}} \\
&= \sqrt{9.99999999999999999999999999851477 \times 10^{-1}} \\
&= 9.99999999999999999999999999925738 4 \times 10^{-1} \\
&= 1 - 7.42616 \times 10^{-28}
\end{aligned}$$
$$\begin{aligned}
\frac{\mathrm{GTD}_1}{\mathrm{GTD}_2} &= 9.99999999999999999999999999777215 \times 10^{-1} \\
&= 1 - 2.22785 \times 10^{-27}
\end{aligned}$$

Time-dilation ratio between System 1 and System 2, computed two ways:

$$\mathrm{GTD}_1/\mathrm{GTD}_2 = 1 - 2.22785 \times 10^{-27} \quad (\text{via } r_s)$$
$$\mathrm{GTD}_1/\mathrm{GTD}_2 = 1 - 2.22785 \times 10^{-27} \quad (\text{via } D_{\mathrm{crit}})$$

$$\text{difference} = 0$$

(below the margin of error imparted by G, 0.00015, the gravitational constant's limit)

This case is tabulated in Table 2, column Classical of Appendix A.

6 Analysis

The reformulation is meant to carry three points, and nothing more ambitious. First, it is a pace-of-time framework that is simply easier to work with: expressed in DTD and compactness, comparing two systems is a direct ratio of geometries, with no benchmark, no subtraction-from-one, and — because D_{crit} cancels — no dependence on G. One returns to GTD, the $\sqrt{1-k}$ form, only at the final step, to convert relative pace into relative clock time. Second, general relativity is always two-system; we merely masquerade it as one. Conventional gravitational time dilation looks like a single-system calculation only because it silently fixes the second system — the same mass at its Schwarzschild radius, where DTD $= 1$. Make that benchmark explicit and the two-system structure, present all along, is plain. Third, treating it as one system is a disservice to the broader mathematics: the collapse to a single system rests on presuming that some property of the hidden second system is a fixed, universal given, a presumption that obscures the real structure and perpetuates a misconception that dilutes the core principle. The honest formulation keeps both systems in view and recognizes the single-system shortcut as a convenience, not as the underlying physics.

7 Conclusions

8 Hypotheses of Explanation

9 Further Exploration

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Conflicts of Interest

The author declares no competing interests.

Data Availability

The complete source materials for this paper, including all LaTeX source files, derivation documents, and revision history, are publicly available at: `https://github.com/denverdata/academic_InertialRelativity_STE_cases`.

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Appendix A Pair Comparison Tables

Tables 1 and 2 present the complete system properties and derived ratios for six system pairs spanning the constraint cases from the derivations in Section 5: static mass (M=M), static radius (R=R), constant density ($\rho = \rho$), constant inertia (I=I), an unconstrained general case, and an STE example. The three algebraic forms of k in Equation 1 produce identical values in every case, alongside the gravitational time-dilation ratio computed independently from the standard Schwarzschild form.

Table 1 System properties for each pair comparison case (six significant figures; GTD in $1-\delta$ form).

	Case 1 (M=M)	Case 2 (R=R)	Case 3 (ρ=ρ)	Case 4 (I=I)	General	Classical
m_1 (kg)	2.00000×10^{30}	1.00000×10^{30}	1.00000×10^{30}	1.00000×10^{32}	1.00000×10^{20}	8.00000×10^{0}
m_2 (kg)	2.00000×10^{30}	1.00000×10^{24}	1.00000×10^{27}	1.00000×10^{30}	1.00000×10^{30}	1.00000×10^{0}
R_1 (m)	3.00000×10^{3}	1.00000×10^{8}	1.00000×10^{8}	1.00000×10^{7}	2.00000×10^{-7}	2.00000×10^{0}
R_2 (m)	7.00000×10^{8}	1.00000×10^{8}	1.00000×10^{7}	1.00000×10^{8}	1.00000×10^{7}	1.00000×10^{0}
ρ_1 (kg/m^3)	1.76839×10^{19}	2.38732×10^{5}	2.38732×10^{5}	2.38732×10^{10}	2.98416×10^{39}	2.38732×10^{-1}
ρ_2 (kg/m^3)	1.39203×10^{3}	2.38732×10^{-1}	2.38732×10^{5}	2.38732×10^{5}	2.38732×10^{8}	2.38732×10^{-1}
I_1 (kg m^2)	7.20000×10^{36}	4.00000×10^{45}	4.00000×10^{45}	4.00000×10^{45}	1.60000×10^{6}	1.28000×10^{1}
I_2 (kg m^2)	3.92000×10^{47}	4.00000×10^{39}	4.00000×10^{40}	4.00000×10^{45}	4.00000×10^{43}	4.00000×10^{-1}
D_1 (kg/m)	6.66667×10^{26}	1.00000×10^{22}	1.00000×10^{22}	1.00000×10^{25}	5.00000×10^{26}	4.00000×10^{0}
D_2 (kg/m)	2.85714×10^{21}	1.00000×10^{16}	1.00000×10^{20}	1.00000×10^{22}	1.00000×10^{23}	1.00000×10^{0}
$r_{s,1}$ (m)	2.97046×10^{3}	1.48523×10^{3}	1.48523×10^{3}	1.48523×10^{5}	1.48523×10^{-7}	1.18819×10^{-26}
$r_{s,2}$ (m)	2.97046×10^{3}	1.48523×10^{-3}	1.48523×10^{0}	1.48523×10^{3}	1.48523×10^{3}	1.48523×10^{-27}
$k_{s,1}$	9.90155×10^{-1}	1.48523×10^{-5}	1.48523×10^{-5}	1.48523×10^{-2}	7.42616×10^{-1}	5.94093×10^{-27}
$k_{s,2}$	4.24352×10^{-6}	1.48523×10^{-11}	1.48523×10^{-7}	1.48523×10^{-5}	1.48523×10^{-4}	1.48523×10^{-27}
$k_{d,1}$	9.90155×10^{-1}	1.48523×10^{-5}	1.48523×10^{-5}	1.48523×10^{-2}	7.42617×10^{-1}	5.94093×10^{-27}
$k_{d,2}$	4.24352×10^{-6}	1.48523×10^{-11}	1.48523×10^{-7}	1.48523×10^{-5}	1.48523×10^{-4}	1.48523×10^{-27}
DTD_1	9.95066×10^{-1}	3.85387×10^{-3}	3.85387×10^{-3}	1.21870×10^{-1}	8.61752×10^{-1}	7.70774×10^{-14}
DTD_2	2.05998×10^{-3}	3.85387×10^{-6}	3.85387×10^{-4}	3.85387×10^{-3}	1.21870×10^{-2}	3.85387×10^{-14}
$\mathrm{GTD}_{s,1}$	$1-9.00777\times10^{-1}$	$1-7.42619\times10^{-6}$	$1-7.42619\times10^{-6}$	$1-7.45394\times10^{-3}$	$1-4.92670\times10^{-1}$	$1-2.97046\times10^{-27}$
$\mathrm{GTD}_{s,2}$	$1-2.12176\times10^{-6}$	$1-7.42616\times10^{-12}$	$1-7.42616\times10^{-8}$	$1-7.42619\times10^{-6}$	$1-7.42644\times10^{-5}$	$1-7.42616\times10^{-28}$
$\mathrm{GTD}_{d,1}$	$1-9.00780\times10^{-1}$	$1-7.42619\times10^{-6}$	$1-7.42619\times10^{-6}$	$1-7.45395\times10^{-3}$	$1-4.92670\times10^{-1}$	$1-2.97047\times10^{-27}$
$\mathrm{GTD}_{d,2}$	$1-2.12176\times10^{-6}$	$1-7.42617\times10^{-12}$	$1-7.42617\times10^{-8}$	$1-7.42619\times10^{-6}$	$1-7.42644\times10^{-5}$	$1-7.42617\times10^{-28}$

Table 2 Derived ratios across constraint cases. Group A is the DTD ratio; Group B the k-ratio; Group C the GTD ratio by both routes; Groups D and E the inertia/density and component analogs.

	Case 1 (M=M)	Case 2 (R=R)	Case 3 (ρ=ρ)	Case 4 (I=I)	General	Classical
Group A — DTD ratio						
$\mathrm{DTD}_1/\mathrm{DTD}_2$	4.83046×10^{2}	1.00000×10^{3}	1.00000×10^{1}	3.16228×10^{1}	7.07107×10^{1}	2.00000×10^{0}
$\sqrt{D_1/D_2}$	4.83046×10^{2}	1.00000×10^{3}	1.00000×10^{1}	3.16228×10^{1}	7.07107×10^{1}	2.00000×10^{0}
$\sqrt{M_1R_2/(M_2R_1)}$	4.83046×10^{2}	1.00000×10^{3}	1.00000×10^{1}	3.16228×10^{1}	7.07107×10^{1}	2.00000×10^{0}
$\sqrt{(I_1/I_2)(R_2/R_1)^3}$	4.83046×10^{2}	1.00000×10^{3}	1.00000×10^{1}	3.16228×10^{1}	7.07107×10^{1}	2.00000×10^{0}
$(I_1/I_2)^{1/5}(\rho_1/\rho_2)^{3/10}$	4.83046×10^{2}	1.00000×10^{3}	1.00000×10^{1}	3.16228×10^{1}	7.07107×10^{1}	2.00000×10^{0}
Group B — k-ratio						
D_1/D_2	2.33333×10^{5}	1.00000×10^{6}	1.00000×10^{2}	1.00000×10^{3}	5.00000×10^{3}	4.00000×10^{0}
$M_1R_2/(M_2R_1)$	2.33333×10^{5}	1.00000×10^{6}	1.00000×10^{2}	1.00000×10^{3}	5.00000×10^{3}	4.00000×10^{0}
$(I_1/I_2)(R_2/R_1)^3$	2.33333×10^{5}	1.00000×10^{6}	1.00000×10^{2}	1.00000×10^{3}	5.00000×10^{3}	4.00000×10^{0}
D_2/D_1	4.28571×10^{-6}	1.00000×10^{-6}	1.00000×10^{-2}	1.00000×10^{-3}	2.00000×10^{-4}	2.50000×10^{-1}
$M_2R_1/(M_1R_2)$	4.28571×10^{-6}	1.00000×10^{-6}	1.00000×10^{-2}	1.00000×10^{-3}	2.00000×10^{-4}	2.50000×10^{-1}
Group C — GTD ratio						
$\mathrm{GTD}_{s,1}/\mathrm{GTD}_{s,2}$	$1-9.00776\times10^{-1}$	$1-7.42618\times10^{-6}$	$1-7.35193\times10^{-6}$	$1-7.44657\times10^{-3}$	$1-4.92632\times10^{-1}$	$1-2.22785\times10^{-27}$
$\mathrm{GTD}_{d,1}/\mathrm{GTD}_{d,2}$	$1-9.00780\times10^{-1}$	$1-7.42619\times10^{-6}$	$1-7.35193\times10^{-6}$	$1-7.44658\times10^{-3}$	$1-4.92633\times10^{-1}$	$1-2.22785\times10^{-27}$
Group D — inertia and density analogs						
I_1/I_2	1.83673×10^{-11}	1.00000×10^{6}	1.00000×10^{5}	1.00000×10^{0}	4.00000×10^{-38}	3.20000×10^{1}
$(I_1/I_2)^{1/2}$	4.28571×10^{-6}	1.00000×10^{3}	3.16228×10^{2}	1.00000×10^{0}	2.00000×10^{-19}	5.65685×10^{0}
$(I_1/I_2)^{1/5}$	7.12540×10^{-3}	1.58489×10^{1}	1.00000×10^{1}	1.00000×10^{0}	3.31445×10^{-8}	2.00000×10^{0}
$(\rho_1/\rho_2)^{1/5}(R_1/R_2)$	7.12540×10^{-3}	1.58489×10^{1}	1.00000×10^{1}	1.00000×10^{0}	3.31445×10^{-8}	2.00000×10^{0}
$(\mathrm{DTD}_1/\mathrm{DTD}_2)^2(R_1/R_2)^3$	1.83673×10^{-11}	1.00000×10^{6}	1.00000×10^{5}	1.00000×10^{0}	4.00000×10^{-38}	3.20000×10^{1}
Group E — component scale factors						
M_1/M_2	1.00000×10^{0}	1.00000×10^{6}	1.00000×10^{3}	1.00000×10^{2}	1.00000×10^{-10}	8.00000×10^{0}
$\sqrt{M_1/M_2}$	1.00000×10^{0}	1.00000×10^{3}	3.16228×10^{1}	1.00000×10^{1}	1.00000×10^{-5}	2.82843×10^{0}
R_1/R_2	4.28571×10^{-6}	1.00000×10^{0}	1.00000×10^{1}	1.00000×10^{-1}	2.00000×10^{-14}	2.00000×10^{0}
R_2/R_1	2.33333×10^{5}	1.00000×10^{0}	1.00000×10^{-1}	1.00000×10^{1}	5.00000×10^{13}	5.00000×10^{-1}
$\sqrt{R_2/R_1}$	4.83046×10^{2}	1.00000×10^{0}	3.16228×10^{-1}	3.16228×10^{0}	7.07107×10^{6}	7.07107×10^{-1}
$\sqrt{M_2^2R_2/(M_1^2R_1)}$	4.83046×10^{2}	1.00000×10^{-6}	3.16228×10^{-4}	3.16228×10^{-2}	7.07107×10^{16}	8.83883×10^{-2}